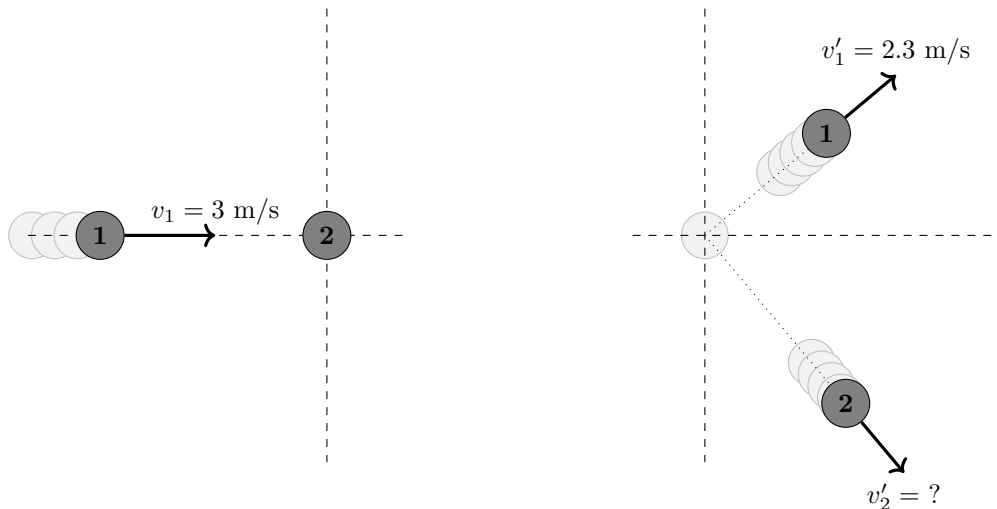


Momentum in Two Dimensions (Consolidation)

1. A 6-kg pool ball moving due east at 3 m/s collides with a second 4-kg ball initially at rest. After the collision, the first ball moves 2.3 m/s at a direction 40° north of east. Find the magnitude and direction of the velocity of the second ball.



Name:

Date:

Period:

2. A truck with a mass of 5000 kg is traveling *north* with a velocity of 13 m/s. At an intersection, it collides with a car ($m = 1500$ kg) traveling *east* with a velocity of 20 m/s. What is the velocity of the vehicles after the collision (before friction has had a chance to take effect)? Assume they stick together after the crash.

